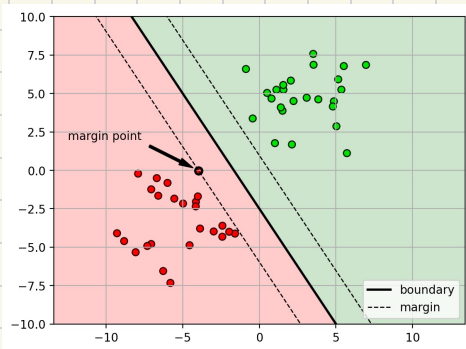


Linear classifier

$$f(x) = w^T x + b$$

$$y \in \begin{cases} +1 & f(x) > 0 \\ -1 & f(x) < 0 \end{cases}$$

Distance from point x
to the boundary is $\frac{|f(x)|}{\|w\|}$

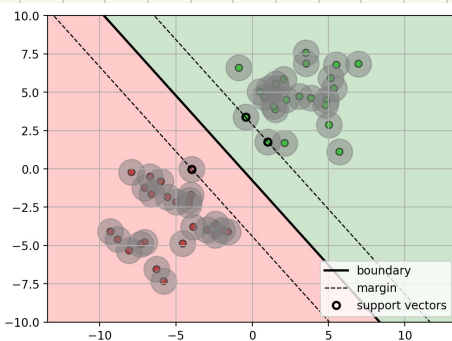
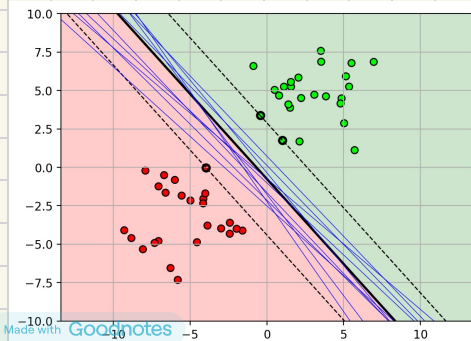


If the data is linearly separable, the distance from the closest point to the boundary is defined as "margin a distance"

$$\gamma = \min_i \frac{|f(x_i)|}{\|w\|} = \min_i \frac{|w^T x_i + b|}{\|w\|} = \min_i \frac{y_i (w^T x_i + b)}{\|w\|}$$

Note: the scale will not affect γ

Idea of SVM: maximize the distance between the boundary and the closest point for better separation

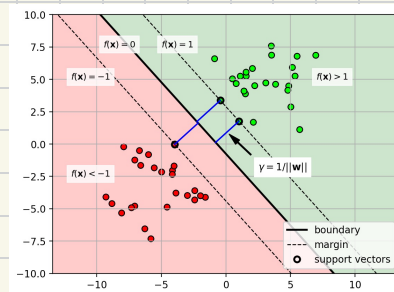


Foundation of SVM

for the nominator:

$$\min_i |w^T x_i + b| = 1$$

objective: $\max_w \frac{1}{\|w\|}$



optimization problem

$$\arg \max_{w, b} \gamma \quad \text{s.t.} \quad \min_i |w^T x_i + b| = 1$$

$$\arg \min_{w, b} \frac{1}{2} \|w\|^2$$

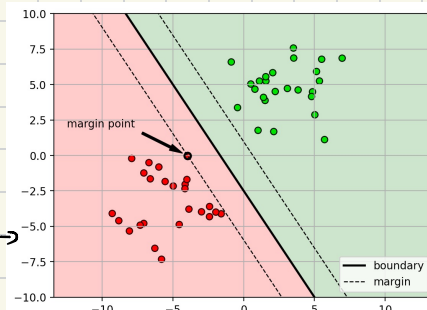
$$|w^T x_i + b| \geq 1$$

At the optimum, the weight w will shrink to make at least one point on the margin

$$\text{SVM: } \arg \min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y_i (w^T x_i + b) \geq 1, \forall i$$

By symmetry, at least one point on each side

to be optimized further



Optimization with inequality constraint

Goal: $\min_x f(x)$ s.t. $g(x) \geq 0$
feasible region (value)

Two possible solution

Solution 1: x^* on the boundary

active equality

at the minimum:

$$\nabla f(x) = \lambda \nabla g(x), \quad \lambda \geq 0 \quad \text{same equation}$$

Both gradients point into feasible region, and cannot decrease $f(x)$ without leaving the boundary

Solution 2: x^* in the feasible region (*inactive equality*)

$$\nabla f(x^*) = 0 \Rightarrow \lambda = 0$$

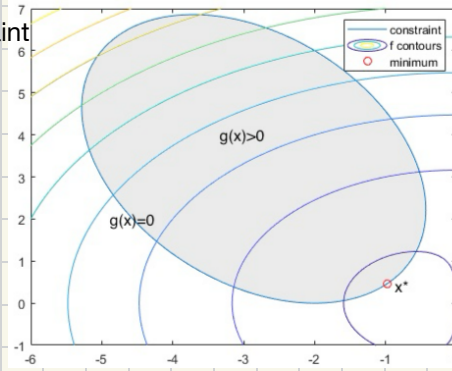
Combining (Lagrange func)

$$L(x, \lambda) = f(x) - \lambda g(x)$$

At the optima $\begin{cases} g(x) \geq 0 \\ \lambda \geq 0 \\ \lambda g(x) = 0 \end{cases}$

KKT Condition

Krause-Kuhn-Tucker



Duality :

Suppose we have optimal λ^* , minimize $L(x, \lambda^*)$

$$L^* = \min_x L(x, \lambda^*) = \min_x [f(x) - \lambda^* g(x)] = f^*$$

Define: $q(\lambda) = \min_x L(x, \lambda) = \min_x [f(x) - \lambda g(x)]$

for every λ , find min of L w.r.t. x

Note:

$$\lambda \geq 0, \quad g(x) \geq 0$$

$$q(\lambda) \leq \min_x f(x) = f(x^*) = f^*$$

maximizing $q(\lambda)$ could obtain $f(x^*)$

Define Dual problem: $q^* = \max_{\lambda \geq 0} q(\lambda)$

Weak duality theorem (duality gap) $q^* \leq f(x^*)$

Strong duality theorem

if 1) $f(x)$ is convex 2) feasible region is convex

3) feasible region is not degenerate

$$\text{Then } q^* = f(x^*) = f^*$$

Solve dual is equivalent to solve the original (primal)

SVM problem: $\arg \min_{w, b, \xi} \frac{1}{2} \|w\|^2$, s.t. $y_i (w^T x_i + b) \geq 1$
 (variable points w, b) $f(w, b)$ $g_i(w, b) \leftrightarrow \alpha_i$

let $\alpha_i \geq 0$ be the langrange multiplier for i -th constraint

Define: $L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^N \alpha_i [y_i (w^T x_i + b) - 1]$
 N different g_i

Dual function: $L(\alpha) = \min_{w, b} L(w, b, \alpha)$

SVM Dual problem: $\max_{\alpha} L(\alpha)$ s.t. $\alpha_i \geq 0, \sum \alpha_i y_i = 0$

$$L(\alpha) = \sum \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$w^* = \sum_i \alpha_i y_i x_i$$

soft margin SVM

margin constraint

$$y_i (w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$$

objective:

$$\min_w \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i$$

$$\text{s.t. } y_i (w^T x_i + b) \geq 1 - \xi_i \quad \forall i$$

$$\xi_i \geq 0 \quad \forall i$$

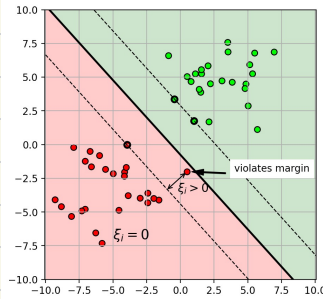
Dual SVM: $\max \sum \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j$

$$\text{s.t. } \sum \alpha_i y_i = 0, 0 \leq \alpha_i \leq C$$

$\alpha_i = 0$ beyond margin

$0 < \alpha_i < C$ on the margin

$\alpha_i = C$ inside margin



The upper constraint of alpha prevent the outlier points from having a large impact on the final classifier weights

primal $\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum \xi_i$

$$\text{s.t. } y_i (w^T x_i + b) \geq 1 - \xi_i \quad \textcircled{1}$$

$$\xi_i \geq 0 \quad \textcircled{2}$$

a) let $\alpha_i \geq 0$ be langrange multiplier for $\textcircled{1}$

$\gamma_i \geq 0$ be Lagrange multiplier for $\textcircled{2}$

Lagrange func:

$$L(w, b, \xi, \alpha, \gamma) = \frac{1}{2} \|w\|^2 + C \sum \xi_i$$

$$- \sum_i \alpha_i (y_i (w^T x_i + b) - 1 + \xi_i) \quad \textcircled{1}$$

$$- \sum_i \gamma_i \xi_i \quad \textcircled{2}$$

b) at the min of L ,

$$\frac{\partial}{\partial w} L = w - \sum_i \alpha_i y_i x_i = 0 \quad \frac{\partial}{\partial b} L = - \sum_i \alpha_i y_i = 0 \quad \frac{\partial}{\partial \xi_i} L = C - \alpha_i - \gamma_i = 0$$

$$w^* = \sum_i \alpha_i y_i x_i$$

$$\sum_i \alpha_i y_i = 0$$

$$\gamma_i + \alpha_i = C$$

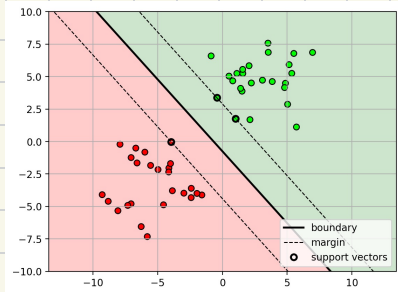
c) $L(\alpha, \gamma) = \min_{w, b, \xi} L(w, b, \xi, \alpha, \gamma)$

$$= \frac{1}{2} \left\| \sum_i \alpha_i y_i x_i \right\|^2 + C \sum \xi_i - \sum_i \alpha_i (y_i (w^T x_i + b) - 1 + \xi_i) - \sum_i \gamma_i \xi_i$$

$$= \frac{1}{2} \left[\left(\sum_i \alpha_i y_i x_i \right)^T \left(\sum_j \alpha_j y_j x_j \right) \right] - \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$+ \sum_i (C - \alpha_i - \gamma_i) \xi_i - \sum_i \alpha_i y_i b + \sum_i \alpha_i$$

$$= \sum \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j = L(\alpha)$$



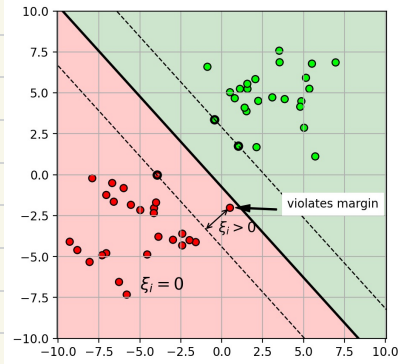
$$\textcircled{1} \quad g(x) = 0 \Rightarrow y_i (w^T x_i + b) - 1 = 0 \\ d_i > 0$$

$W^* = \sum d_i y_i x_i$ only depends on
the points on margin
(support vector)

$$\textcircled{2} \quad g(x) > 0 \Rightarrow d_i = 0 \\ y_i (w^T x_i + b) > 1$$

At the optima

$$\begin{cases} g(x) \geq 0 \Rightarrow g_i(w, b) \geq 0 \\ \lambda \geq 0 \Rightarrow d_i \geq 0 \\ \lambda g(x) = 0 \end{cases}$$



$$y_i (w^T x_i + b) \geq 1 - \xi_i \quad 0 \leq d_i \leq C \\ \xi_i \geq 0 \quad 0 \leq \gamma_i \leq C$$

$$\gamma_i > 0 \begin{cases} d_i = 0, \xi_i = 0 \Rightarrow y_i (w^T x_i + b) > 1 \\ 0 < d_i < C, \xi_i = 0 \Rightarrow y_i (w^T x_i + b) = 1 \end{cases}$$

$$\gamma_i = 0 \Rightarrow \begin{cases} d_i = C \\ \xi_i > 0 \Rightarrow y_i (w^T x_i + b) = 1 - \xi_i \end{cases}$$